Prasanna Gayathri T S

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Software Engineering Student specializing in Machine Learning, AI Automation, and Production-Grade ML Pipelines



SUMMARY

Integrated MTech Software Engineering student (CGPA 8.57) with hands-on experience building API-driven applications, NLP/ML pipelines, LLM-powered applications (Gemini API, RAG) and deployed web apps. Skilled in Python and Java, REST API integration, Git-based workflows and cloud deployment, with a security-focused internship on an Identity & Access Management (IAM) dashboard. Eager to build reusable AI automation and agentic workflows that scale across business teams.

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, JavaScript, Android (Java)
AI/ML & Automation: Generative AI & LLM Integration (Gemini API), Prompt Engineering, RAG, Machine Learning, NLP, BERT (fine-tuning), ResNet50, OCR (EasyOCR), TensorFlow, Scikit-learn, NumPy, Pandas, Statistical Analysis, ML Pipeline Design, Workflow Automation
APIs & Integration: RESTful APIs, LLM & Third-Party API Integration (Gemini, Open Food Facts), Retry & Error Handling, Asynchronous Data Fetching, Node.js, Client–Server Integration
Cloud, Deployment & Engineering Practices: Git/GitHub (version control), Cloud Hosting, Modular & Reusable Pipeline Design, Google Colab
Security & Identity: IAM Dashboard Development, Secure Authentication, Role-Based Access Control, Session Management, Vulnerability Tracking
Data & Tools: MySQL, MongoDB, Apache Hive, Hadoop, Microsoft Excel, Power BI, Jira, Firebase, Figma, VS Code
Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems
Soft Skills: Analytical Thinking, Problem Solving, Communication, Cross-Team Collaboration

WORK EXPERIENCE

Hyntech Business Solutions, Chennai (Remote) 1 Jul 2025 – 28 Jul 2025
Frontend Developer Intern

- Built a responsive Identity & Access Management (IAM) dashboard in JavaScript with secure authentication and dynamic data rendering to track IP and domain vulnerabilities in real time, reducing manual monitoring by 35–40%.
- Integrated RESTful APIs for seamless client–server communication and asynchronous data fetching, powering real-time updates across 3+ dashboard modules.
- Collaborated with the team using Git version control and Jira to deliver features in a structured workflow.

PROJECTS

SafeScan — LLM-Driven Multimodal Product Recognition with Allergen Detection *(Python, TensorFlow, ResNet50, EasyOCR, Gemini API, RAG, Open Food Facts)*

- Built a two-stage pipeline: ResNet50 image embeddings fused with TF-IDF OCR text features feed a dense neural network that classifies ~11,900 grocery product images into 21 categories, reaching 88.74% test accuracy.
- Integrated the Gemini API as an LLM reasoning layer with engineered prompts to detect allergen conflicts against a user allergy profile from noisy OCR ingredient text, returning a structured Safe / Conflict / Uncertain verdict.
- Grounded LLM outputs with RAG over the Open Food Facts API and a deterministic allergen-ontology check, and highlighted conflicting ingredients on the product image using OCR bounding boxes.
- Computed health grades with the published Nutri-Score algorithm (LLM used only for nutrient extraction) and added Mondrian conformal prediction for calibrated 90% confidence sets.
- Hardened for production: exponential-backoff retries, separate handling of 401 / 429 / 404 / 503 API errors, and graceful degradation so a failed LLM call still returns classification and ontology results.

Amazon Sustainability Classifier — Large-Scale NLP Pipeline *(Python, Hadoop, BERT, NLP, Scikit-learn)*

- Built a distributed text-processing pipeline on Hadoop using MapReduce to batch-process 1,000+ Amazon customer reviews on commodity hardware.
- Automated data wrangling and preprocessing with Pandas and NumPy (tokenization, null handling, feature transformation) to prepare model-ready inputs.
- Fine-tuned BERT embeddings for sentiment analysis (positive / negative / neutral), improving precision over a TF-IDF baseline.
- Designed a multi-label classifier tagging products as sustainable or non-sustainable, covering the full pipeline from raw data to actionable insight.

Paw Connect — Pet Care Service & Community Platform *(HTML, CSS, Node.js, MongoDB, REST API, GitHub)*

- Delivered the platform from development to production using GitHub version control and cloud hosting, validating stability across booking, community feed and messaging modules.
- Built a Node.js and MongoDB service layer handling concurrent transactions, session management and role-based data access.

EDUCATION

Vellore Institute of Technology, Vellore — MTech Software Engineering (Integrated) | CGPA: 8.57 2022 – 2027
Sunbeam CBSE, Katpadi — Grade 12: 79.2% | Grade 10: 84% Jun 2020 – 2022

CERTIFICATIONS

- Software Engineering Job Simulation – Accenture (Forage)
- Data Science and Analytics – HP